

A Step-by-step production and assembly guide for

UPick

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UPick

PRODUCT DEFINITIONS

WHO?

The potential users include individuals who use wheelchairs, people with neurological conditions or disabilities that affect hand dexterity, grip strength, or fine motor skills, and elderly individuals or those with age-related mobility limitations.

WHY?

UPick provides a practical, innovative solution tailored to individuals with limited hand mobility needs, promotes autonomy and boosts self-confidence, enabling users to feel more capable and in control of their daily lives.

HOW?

This innovative development will ensure the maintenance of the users independence and enhance their ability to function autonomously.

WHAT?

An innovative electronic product, operating on vacuum technology and rechargeable, which helps people with disabilities lift objects from the floor through adhesion.

[link - watch video](#)



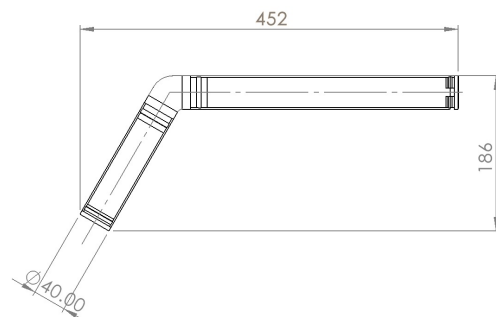
PRICING + SPECIFICATIONS

ESTIMATED COST

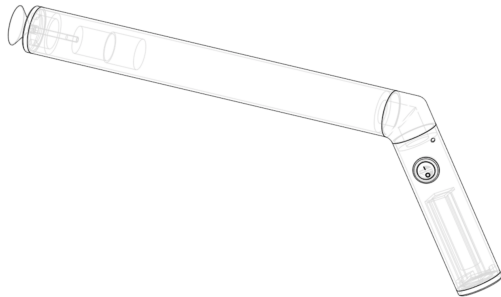
Materials	25\$
Machine Time	4 Hrs
Assembly Time	4 Hrs

DIMENSIONS

Height	186Millimeters
Width	40Millimeters
Length	452Millimeters



PRODUCT VIEWS





BEFORE YOU START



Carefully read the entire product file to ensure that you have all the required tools, materials and correct printer settings **BEFORE** you start making the product.



Always wear protective gear when working with power tools.



Send us feedback if you are having difficulty building this product - hello@tomglobal.org



Have fun and give yourself a pat on the back for making the world a better place!



TOOLS NEEDED

- Protective glasses
- Protective headphones
- PC or MAC I7 +
- Prusa slicer
- Measurement tool
- Clamps
- Pliers
- Needle-nose pliers
- Cutter
- Pull saw
- Electric Srewdriver
- Dremel
- FDM Printer
- Soldering set
- Voltmeter
- Cable stripper

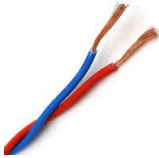
Bill Of Materials

Centimeters to Inches Converter

Item #	Image	Description	Material	Manufacturing instruction (Technology, printing infill etc.)	Dimensions	QTY	Purchase Link/Supplier
1		PVC tube	PVC	of the shelf	Diameter 40/36 mm, length 1:14cm length 2: 33 cm	2	Purchase Link
2		Mini 370 air pump 3.7v	electronic	Diaphragm, 280 3V Mini Air Peristaltic Pump, DC for Pumping Environments	000 cm	1	Purchase Link
3		3.7V Lithium battery	Lithium	3.7 Volt 18650 Rechargeable Battery, 3500mAh Battery Lithium Battery for Flashlight, Headlamp, Flat Top Battery	000 cm	1	Purchase Link
4		suction pad and adapter	VP25 SI and IM21	of the shelf	D=25 ml	2	Purchase Link
5		Power switch	plastic	Rocker Switch - On/Off/On (Round) COM-14978 RoHS	000 cm	1	Purchase Link

Bill Of Materials

Centimeters to Inches Converter

Item #	Image	Description	Material	Manufacturing instruction (Technology, printing infill etc.)	Dimensions	QTY	Purchase Link/Supplier
6		Electrical wiring	Low Voltage Cable 16AWG	Electrical Wire 2 Conductors Cable Flexible, Low Voltage	1 m	1	Purchase Link
7		Battery housing-18650 Bat	plastic	18650 Battery Charger, Suitable for 3.7v Lithium Batteries	000 cm	1	Purchase Link
8	 TP4056 Type-C USB Without Current Protection	Charging Module	Mini Type C Micro USB TP 4056 1A	HiLetgo 3pcs TP4056 Type-c USB 5V 1A 18650 Lithium Battery Charger Module Charging Board with Dual Protection Functions	000 cm	1	Purchase Link
9		3d connectors and plugs	3D printed	petg	000 cm	3	Purchase Link
10		LED lights 5mm (optional)	led	5mm LED Light Emitting Diode Diffused Green	000 cm	3	Purchase Link



Bill Of Materials

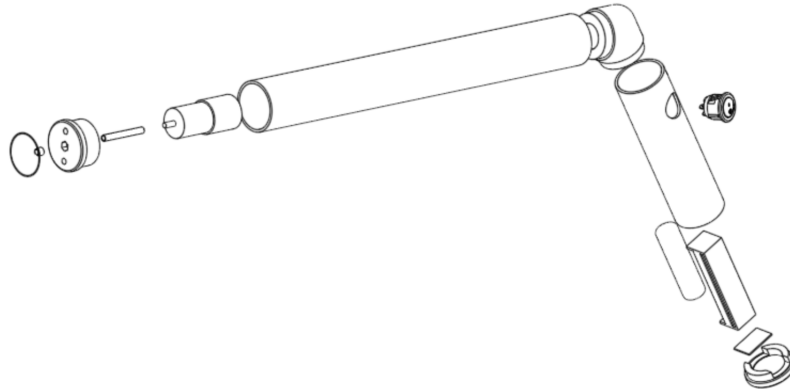
Centimeters to Inches Converter

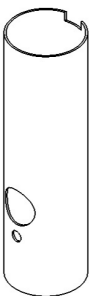
Item #	Image	Description	Material	Manufacturing instruction (Technology, printing infill etc.)	Dimensions	QTY	Purchase Link/Supplier
11		Pump Extension + Hex Nut	TPU Filament, Hex nut	3D print with flexible material (TPU), insert M5 8 mm hex nut to place	8 mm	1	Purchase Link



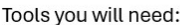
UPick

EXPLODED VIEW





Holes in tubes diagram



Electrical circuit

STEP #1

Cut the tubes

Tip: Cut the main tube to the required length for the body: 33cm. Cut a shorter piece for the handle section: 14cm

Tools:

Protective glasses

Measurement tool

Clamps

Cutter

Dremel



STEP #2

Drill holes in handle

Tip: Create a hole for the power switch Create a hole for the LED light.

Tools:

Protective glasses

Measurement tool

Clamps

Cutter

Dremel

STEP #3

Cut Charger hole

Tip: Create a hole for the charging port, using a saw - make sure it fits the printed part

Tools:

Protective glasses
Measurement tool
Clamps
Cutter
Pull saw
Dremel



STEP #4

Sand Tubes

Tip: You should sand the tubes with a fine sanding paper (500 GRIT minimum)

Tools:

Protective glasses
Measurement tool
Clamps
Cutter
Pull saw
Dremel

STEP #5

Paint the Tubes

Tip: Outside, spray paint the tubes in at least three coats, waiting an hour in between each for the paint to dry. In between each layer: re-sand. Then paint with at least two coats of finish, also with an hour between each. This is easiest if you stand the tubes up on a wooden stick.

Tools:

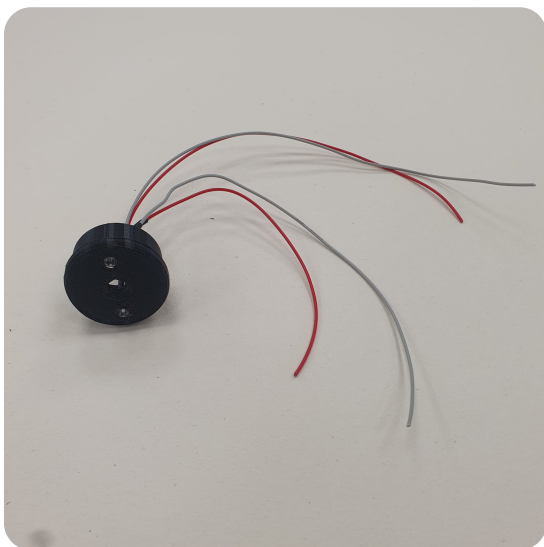


STEP #6

Connect the LEDs

Tip: Connect the LEDs to the electronic wires and insert them into the designated part in the cover

Tools:

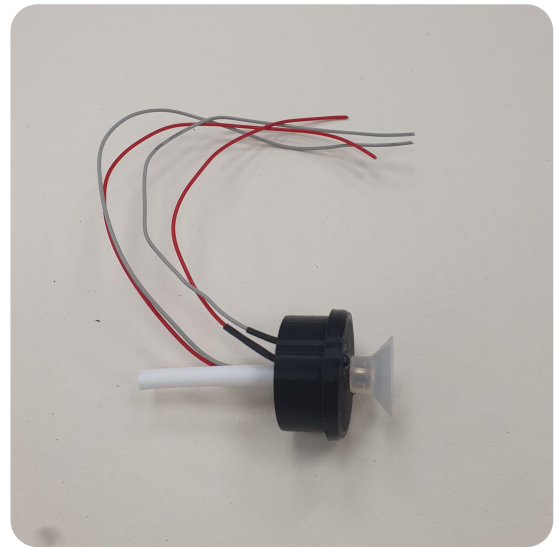


STEP #7

Attach the suction cup

Tip: Connect the vacuum nozzle to the plastic tube and insert it into the bottom cover

Tools:

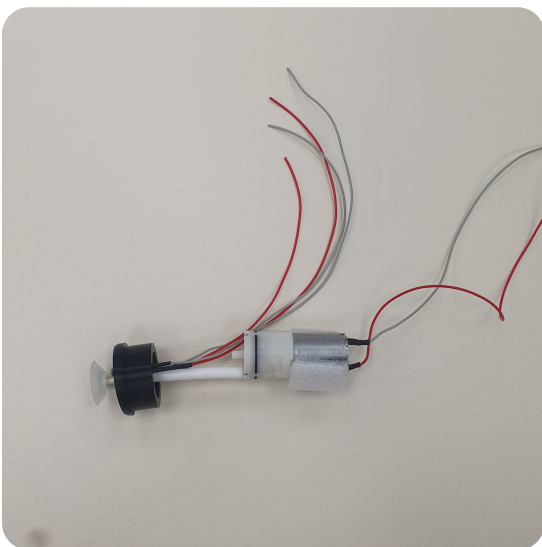


STEP #8

Connect the air pump

Tip: Connect the tube that attaches to the vacuum nozzle to the pump, and secure it using heat

Tools:



STEP #9

Make the Circuit

Tip: We recommend taping the circuit before soldering it to make sure you follow the drawing and things are working. At this point - check for the need in resistors based on the LEDs you used, we didn't use resistors for 1 green + 2 blue LEDs.

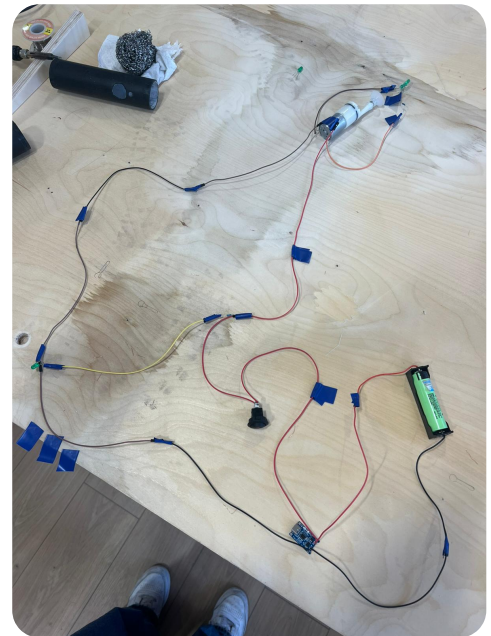
Tools:

Needle-nose pliers

Cutter

Voltmeter

Cable stripper



STEP #10

Soldering the Circuit

Tip: Solder the circuit carefully. Leave enough length to go through both tubes and ensure the pump and battery both have room to slide in and out. The switch needs to be inserted from the outside, so keep that in mind.

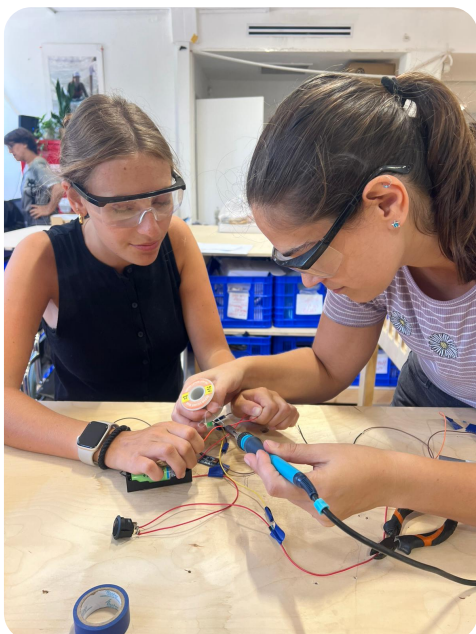
Tools:

Needle-nose pliers

Cutter

Voltmeter

Cable stripper



STEP #11

Connect the two tube

Tip: Use the 3D printed connectors to join the handle and main body securely. Close and tighten the upper and lower covers. If necessary, use glue or screws.

Tools:

PC or MAC I7 +
FDM Printer



STEP #12

Quality check

Tip: Ensure all connections are secure Test the vacuum lifting capability Verify the charging adapter fits and functions correctly

Tools:

PC or MAC I7 +
FDM Printer



THANK YOU FOR
MAKING WITH US
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Feel free to send us your feedback
to us at hello@tomglobal.org

